

ZONAL E/E ARCHITECTURE COMPARATIVE STUDY REPORT

Legacy Point-to-Point vs 4-Zone Ethernet Architecture

Fleet Scope:

Freightliner Cascadia 126 (2020) | Western Star 49X (2021) | Mercedes-Benz Sprinter 519 CDI (2021)

Total ECUs modelled: 51 | Total connections modelled: 60

Tool: Zonal E/E Architecture Analyzer | AI Review: Groq Llama 3.3 70B

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Primary data source: DTNA SS-1033423, NHTSA public database

How to Read This Report

This report compares two ways of wiring electronics inside a commercial truck. The **old way (Legacy / Point-to-Point)** connects every ECU directly to every other ECU it talks to using individual wires. The **new way (Zonal Architecture)** divides the truck into 4 physical regions. Each ECU connects to a nearby hub (Zone Controller) with a short wire. The 4 hubs talk to each other over one Ethernet cable running along the truck. This tool models 3 vehicles, calculates the difference in wire length, weight, and complexity, checks if zone boundaries are optimal, and computes a recommended zone design balancing wire length and communication efficiency. A full abbreviation guide is on the next page.

Abbreviations and Full Forms

Every shortform used in this report is listed below.

Abbreviation	Full Form	Abbreviation	Full Form
ABS	Anti-lock Braking System	J1939	SAE Standard for heavy-duty vehicle CAN network
ACM	Aftertreatment Control Module	LDW	Lane Departure Warning
BCM	Body Control Module	MCM	Motor Control Module
CAN	Controller Area Network	MPC	Multi-Purpose Camera
CBZ	Cab Zone	MSF	Modular Switch Field
CGW	Central Gateway	NHTSA	National Highway Traffic Safety Administration
CHZ	Chassis Zone	ONGUARD	OnGuard Headway / Collision Controller
CPC	Common Powertrain Controller	OTA	Over-the-Air software update
CTP	Common Telematics Platform	PTZ	Powertrain Zone
DCMD	Door Controller — Driver Side	SAE	Society of Automotive Engineers
DCMP	Door Controller — Passenger Side	SAM_CAB	Signal Acquisition Module — Cab
DEF	Diesel Exhaust Fluid	SAM_CH	Signal Acquisition Module — Chassis
DPF	Diesel Particulate Filter	SAS	Steering Angle Sensor
DTCI	Daimler Truck Innovation Center India	SCR	Selective Catalytic Reduction
DTNA	Daimler Trucks North America	SRS	Supplemental Restraint System (Airbag)
E/E	Electrical / Electronic	TCM	Transmission Control Module
ECAS	Electronic Controlled Air Suspension	TCO	Tachograph
ECM	Engine Control Module	TLM	Telematics Module
ECU	Electronic Control Unit	TPMS	Tire Pressure Monitoring System
EPS	Electric Power Steering	TSN	Time-Sensitive Networking
ESC	Electronic Stability Control	ZC	Zone Controller
FCM	Forward Collision Module	ZC-CAB	Zone Controller — Cab Zone
FCU	Front Climate Control Unit	ZC-CH	Zone Controller — Chassis Zone
FRZ	Front Zone	ZC-FR	Zone Controller — Front Zone
HVAC	Heating Ventilation and Air Conditioning	ZC-PT	Zone Controller — Powertrain Zone
ICU	Instrument Cluster Unit	100BASE-T1	Automotive Ethernet — single-pair 100 Mbps

1. Executive Summary

The zonal E/E architecture analysis of the Freightliner Cascadia 126 (2020) reveals a 53.7% reduction in wire length, from 52.5m to 24.3m, with a zone efficiency score of 7.0%. The most important finding is the significant wire reduction, which translates to a weight saving of 3.4 kg. The ICU emerges as a critical communication hub with 12 connections, highlighting its potential as a single point of failure.

1.1 Key Metrics — All Three Vehicles

Metric	Freightliner Cascadia 126	Western Star 49X	MB Sprinter 519 CDI
ECU Count	22	17	12
Legacy Wire Length	52.5m	56.5m	20.5m
Zonal Wire Length	24.3m	19.8m	13.9m
Wire Length Saved	53.7%	65.0%	32.2%
Cross-Zone Runs: Legacy	16	16	8
Cross-Zone Runs: Zonal	3	3	3
Harness Weight Saved	3.4kg	4.4kg	0.8kg
Zone Efficiency Score	7.0%	-4.9%	31.2%

Table 1: Key metrics. Green = improvement. Red = legacy (higher is worse).

2. Background — What Is Zonal Architecture?

A modern truck has dozens of ECUs — small computers controlling everything from the engine and brakes to the dashboard and door locks.

The Old Way — Legacy Point-to-Point Architecture

If ECU A needs to talk to ECU B, a dedicated wire is run between them. With 22 ECUs and 25 communication pairs on a Cascadia 126, this creates a dense web of wires crossing the full truck. The engine controller (MCM) talking to the dashboard (ICU) needs a wire running ~5.5 meters. Result: 52.5m of wire and 6.3 kg of copper.

The New Way — 4-Zone Ethernet Architecture

The truck is divided into 4 physical regions. Each ECU connects only to a nearby Zone Controller with a short wire (0.5–1.5m). The 4 Zone Controllers communicate over one Ethernet cable (100BASE-T1) running along the truck spine. Result: 24.3m total wire, 2.9 kg, and only 3 cross-vehicle runs instead of 16.

Feature	Legacy Architecture	Zonal Architecture
Wiring principle	Direct wire between every communicating pair	Each ECU to local Zone Controller to Ethernet backbone
Cross-truck runs	One per communication pair (up to 16 long runs)	Only 3 backbone segments total
Wire length (Cascadia)	52.5 m	24.3 m (-53.7%)
Fault isolation	One broken wire can affect the whole system	Fault stays within the affected zone
Scalability	Complexity grows fast with each new ECU	Each new ECU adds only one short local wire
Industry status	Current standard on most trucks in service	Target for next-generation truck platforms

Table 2: Legacy vs zonal architecture comparison.

3. Vehicle Configurations and Zone Layouts

Three vehicles were modelled. ECUs were assigned to zones based on actual mounting location, cross-referenced with official DTNA and Mercedes-Benz documentation.

3.1 Freightliner Cascadia 126

Total ECUs: 22 | **Connections:** 25 | **Source:** DTNA SS-1033423, NHTSA public database

Zone	ID	Zone Controller	ECUs in Zone	Count
Powertrain Zone	PTZ	ZC-PT	MCM, ACM, CPC, TCM	4
Chassis Zone	CHZ	ZC-CH	ABS, ECAS, TPMS, SAM_CHASSIS	4
Cab Zone	CBZ	ZC-CAB	ICU, SAM_CAB, CGW, FCU, SRS, MSF, PARKSMART, DCMD, DCOMP, TCO	10
Front Zone	FRZ	ZC-FR	MPC, ONGUARD, SAS, CTP	4

Table 3.1: Zone layout — Freightliner Cascadia 126

3.2 Western Star 49X

Total ECUs: 17 | **Connections:** 22 | **Source:** DTNA SS-1033423, Western Star Bodybuilder Manual Rev 3.1

Zone	ID	Zone Controller	ECUs in Zone	Count
Powertrain Zone	PTZ	ZC-PT	MCM, ACM, CPC, TCM, PTO	5
Chassis Zone	CHZ	ZC-CH	ABS, AXLE, TPMS, SAM_CH	4
Cab Zone	CBZ	ZC-CAB	ICU, CGW, SAM_CAB, FCU, MSF	5
Front Zone	FRZ	ZC-FR	ONGUARD, SAS, CTP	3

Table 3.2: Zone layout — Western Star 49X

3.3 Mercedes-Benz Sprinter 519 CDI

Total ECUs: 12 | **Connections:** 13 | **Source:** Mercedes-Benz XENTRY public diagnostic documentation

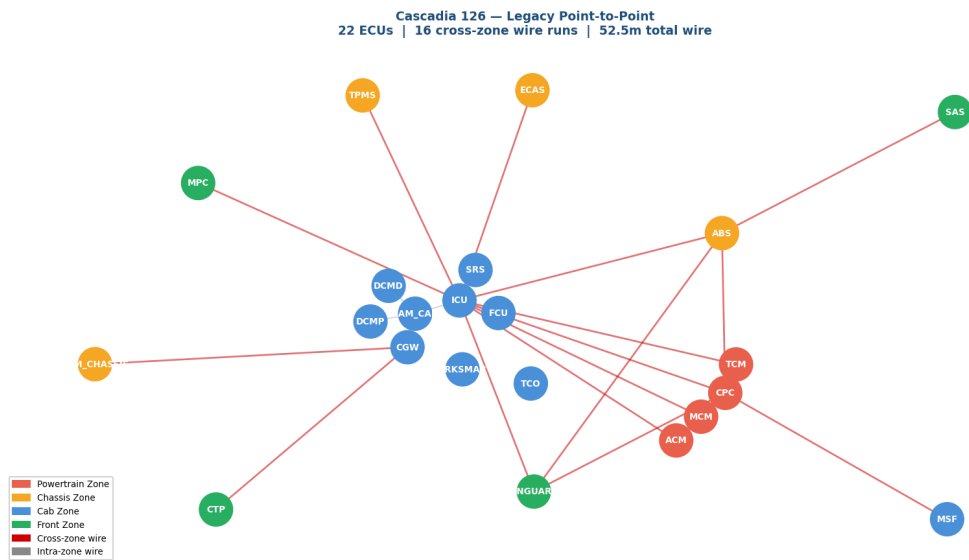
Zone	ID	Zone Controller	ECUs in Zone	Count
Powertrain Zone	PTZ	ZC-PT	ECM, TCU	2
Chassis Zone	CHZ	ZC-CH	ABS_ESP, EPS	2
Cab Zone	CBZ	ZC-CAB	IC, BCM, HVAC, ACM_AIR, TCO	5
Front Zone	FRZ	ZC-FR	FCM, LDW, TLM	3

Table 3.3: Zone layout — Mercedes-Benz Sprinter 519 CDI

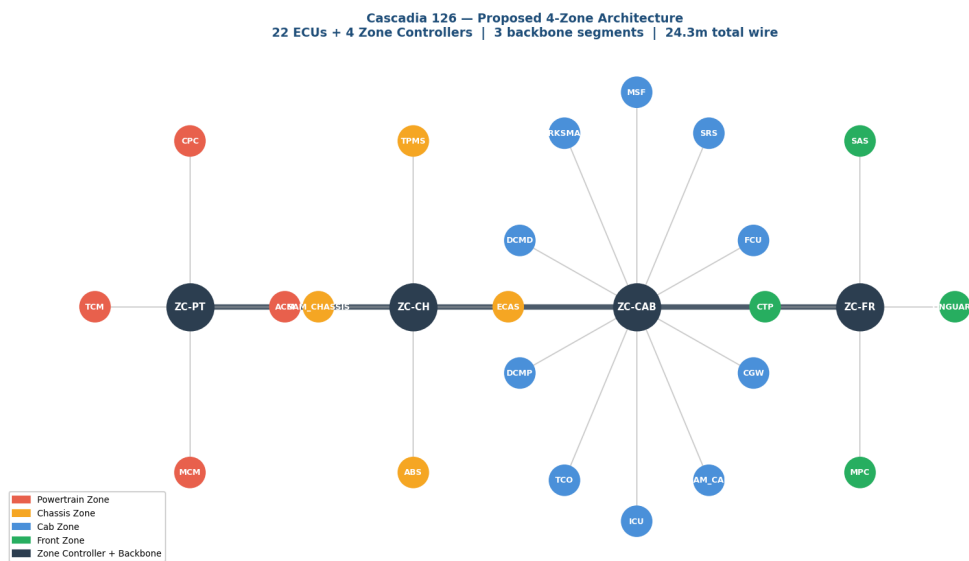
4. Network Topology Diagrams — All Three Vehicles

Each vehicle shown in legacy layout (red = cross-zone wires) and proposed 4-zone zonal layout (dark hubs = Zone Controllers).

4.1 Freightliner Cascadia 126

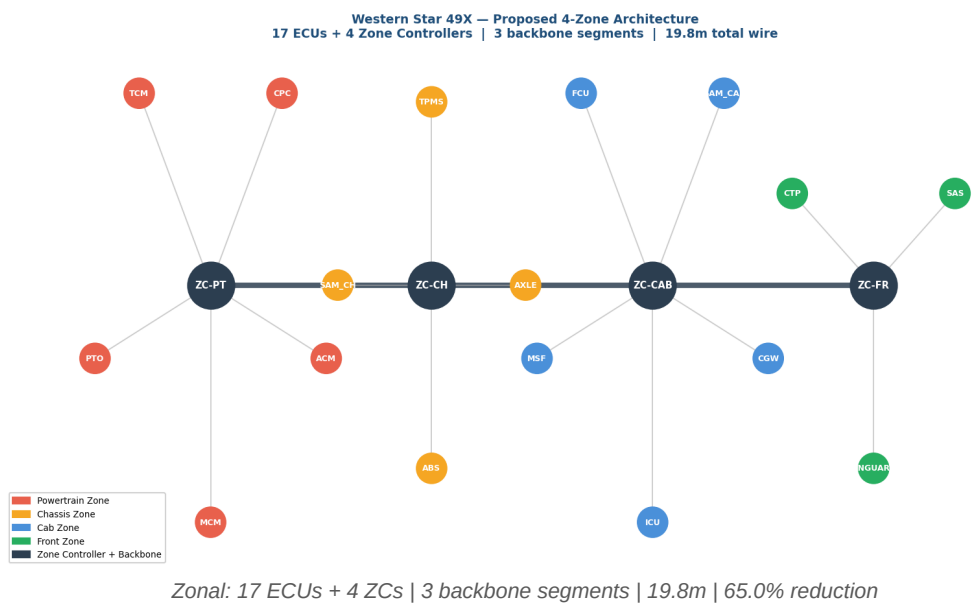
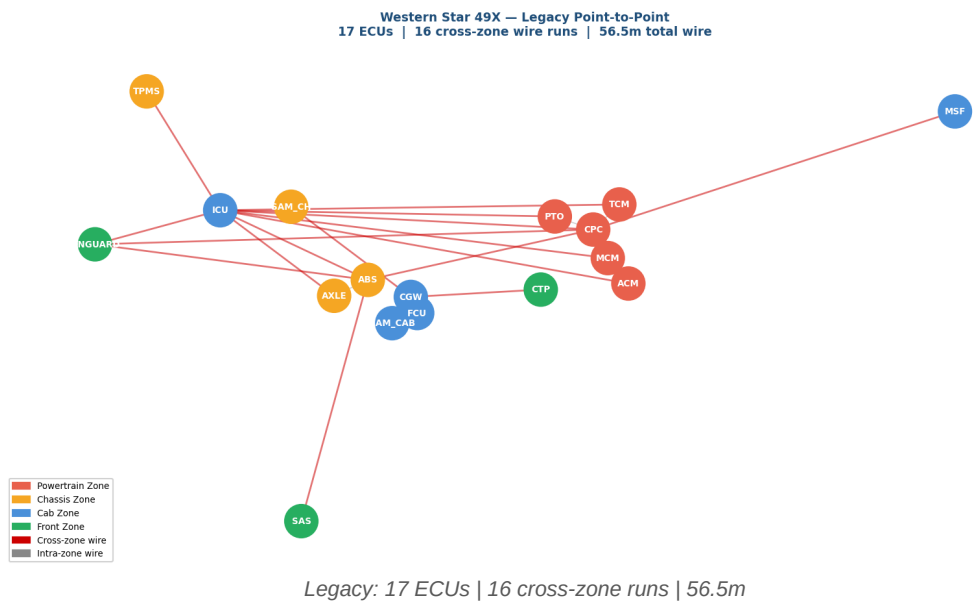


Legacy: 22 ECUs | 16 cross-zone runs | 52.5m

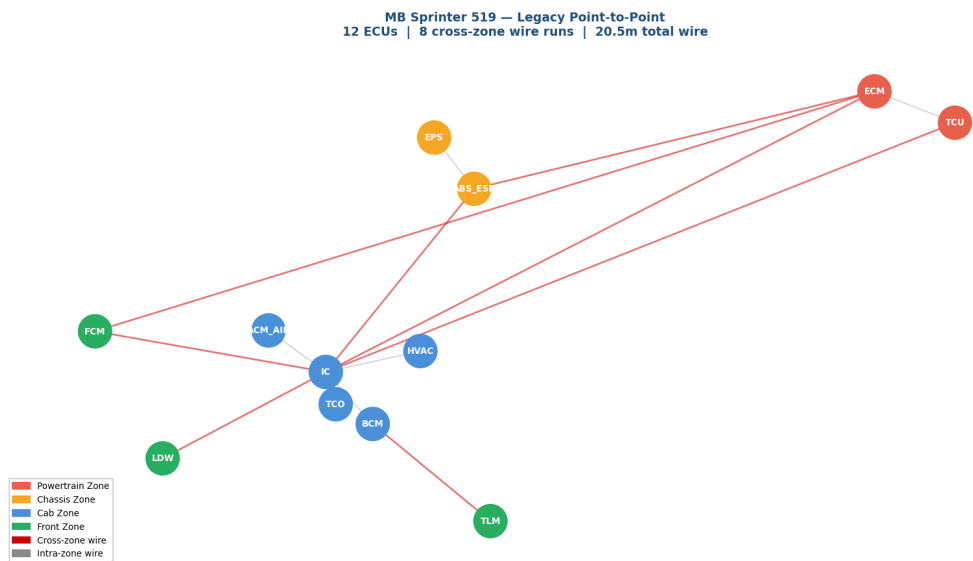


Zonal: 22 ECUs + 4 ZCs | 3 backbone segments | 24.3m | 53.7% reduction

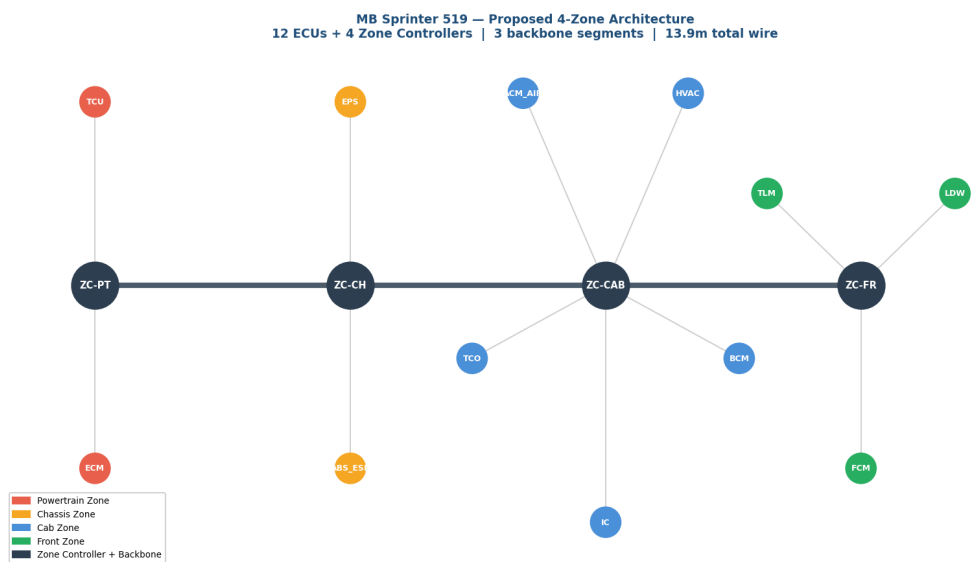
4.2 Western Star 49X



4.3 Mercedes-Benz Sprinter 519 CDI



Legacy: 12 ECUs | 8 cross-zone runs | 20.5m



Zonal: 12 ECUs + 4 ZCs | 3 backbone segments | 13.9m | 32.2% reduction

5. Zone Assignment Optimization — Three-Way Comparison

This section answers the question: **what is the best possible zone design?**

The tool computes three zone assignment strategies and compares them side by side. **Option A** is the current industry design — ECUs grouped by physical location. **Option C** is the theoretical communication-optimal design — ECUs grouped purely by who they talk to, ignoring physical constraints. This is not implementable in practice because some ECUs would need to connect to a Zone Controller far away, defeating the purpose of short local stubs. **Option B** is the recommended balanced design — it only moves ECUs where (1) the communication benefit is clear AND (2) the wire cost stays within adjacent-zone range (3.0m or less). This is the practically implementable optimum.

Why do zones need optimizing at all? Zones are currently drawn based on physical location — where each ECU is bolted on the truck. But communication patterns do not follow geography. The engine controller (MCM) talks constantly to the dashboard (ICU) even though they are on opposite ends of the truck. A graph algorithm called Greedy Modularity Community Detection confirms this — it finds that the mathematically optimal zone groupings based purely on communication patterns produce a completely different layout from the physical zones (7% match on the Cascadia, confirming the tradeoff is structural). The top communication hubs — ICU with 12 connections, CPC with 6, ABS with 4 — all communicate heavily across zone boundaries, which is exactly why the current physical zone design produces 16 cross-zone runs. Option B reduces this to 7 by moving ECUs to zones where most of their communication partners already live, while keeping all wire stubs within the adjacent-zone distance limit.

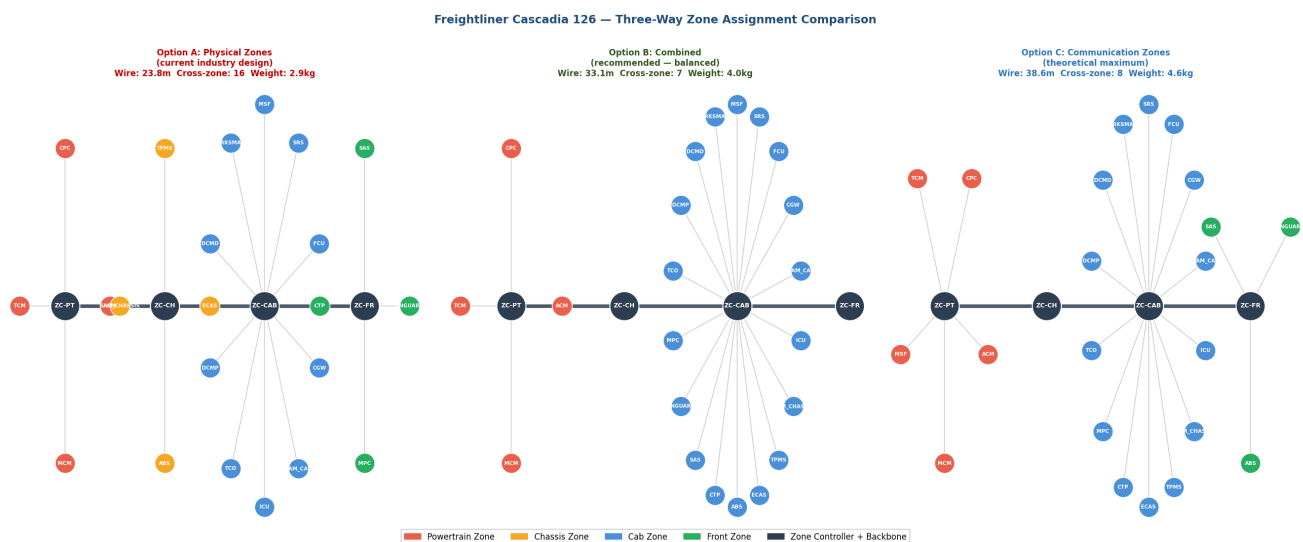


Figure D: Three-way zone assignment comparison. Node colour shows the ASSIGNED zone (not physical mounting zone). Option B ECUs that moved appear in a different colour from their physical location.

Metric	Option A Physical (current)	Option B Combined (recommended)	Option C Communication (theoretical)
Total wire length	23.8 m	33.1 m	38.6 m
Cross-zone runs	16	7	8
Est. harness weight	2.9 kg	4.0 kg	4.6 kg
Physically feasible	YES	YES	NO — some ECUs too far from ZC

Table D: Three-way comparison. Option B (green) is the recommended implementable design.

5.1 Recommended Zone Changes

The optimizer recommends moving 8 ECUs from their physical zone. All moves keep wire cost within 3.0m (adjacent zone range) while significantly reducing cross-zone traffic from 16 to 7 runs.

ECU	Current Zone	Recommended Zone	Reason
ABS	Chassis Zone	Cab Zone	3/4 connections in new zone vs 0/4 in current zone
ECAS	Chassis Zone	Cab Zone	1/1 connections in new zone vs 0/1 in current zone
TPMS	Chassis Zone	Cab Zone	1/1 connections in new zone vs 0/1 in current zone
SAM_CHASSIS	Chassis Zone	Cab Zone	1/1 connections in new zone vs 0/1 in current zone
MPC	Front Zone	Cab Zone	1/1 connections in new zone vs 0/1 in current zone
ONGUARD	Front Zone	Cab Zone	2/3 connections in new zone vs 0/3 in current zone
SAS	Front Zone	Cab Zone	1/1 connections in new zone vs 0/1 in current zone
CTP	Front Zone	Cab Zone	1/1 connections in new zone vs 0/1 in current zone

Table E: ECU zone reassignment recommendations. All moves are physically feasible.

5.2 Pareto Tradeoff Curve

The chart below shows the tradeoff between wire length and communication efficiency. Moving from Option A to C reduces cross-zone runs but increases wire length. Option B sits at the knee of the curve — maximum communication benefit for minimum additional wire cost.

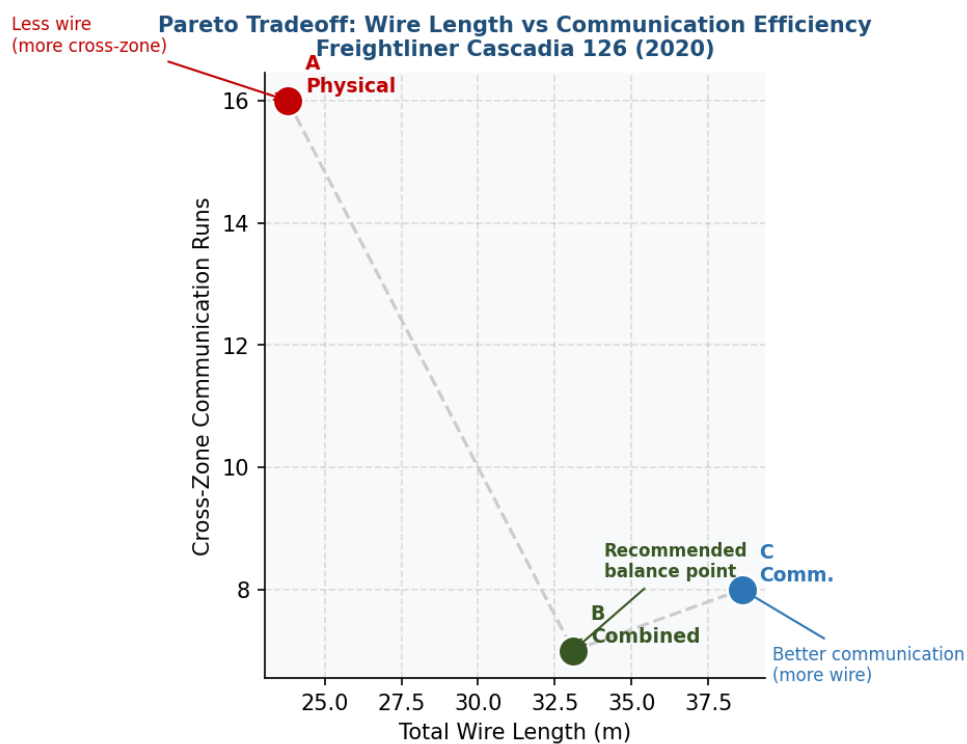


Figure E: Pareto curve. Option B is the recommended operating point — it achieves most of the communication gain at a fraction of the wire cost penalty.

6. Harness Reduction Analysis

Wire length, cross-zone run count, and weight reduction for each vehicle.

6.1 Freightliner Cascadia 126

Metric	Legacy (Point-to-Point)	Zonal (4-Zone)	Reduction
Total wire length	52.5 m	24.3 m	53.7%
Cross-zone wire runs	16	3	81.2%
Est. harness weight	6.3 kg	2.9 kg	3.4 kg saved

Table 6.1: Harness metrics — Freightliner Cascadia 126

6.2 Western Star 49X

Metric	Legacy (Point-to-Point)	Zonal (4-Zone)	Reduction
Total wire length	56.5 m	19.8 m	65.0%
Cross-zone wire runs	16	3	81.2%
Est. harness weight	6.8 kg	2.4 kg	4.4 kg saved

Table 6.2: Harness metrics — Western Star 49X

6.3 Mercedes-Benz Sprinter 519 CDI

Metric	Legacy (Point-to-Point)	Zonal (4-Zone)	Reduction
Total wire length	20.5 m	13.9 m	32.2%
Cross-zone wire runs	8	3	62.5%
Est. harness weight	2.5 kg	1.7 kg	0.8 kg saved

Table 6.3: Harness metrics — Mercedes-Benz Sprinter 519 CDI

The reduction in wire length from 52.5m to 24.3m is driven by the zonal architecture, which minimizes cross-zone runs. The number of cross-zone runs decreases from 16 to 3, indicating a substantial reduction in inter-zone connections. Specifically, the Powertrain Zone (PTZ) and Chassis Zone (CHZ) likely contribute to this reduction, as they have fewer connections to other zones. The elimination of redundant connections between zones, such as those between the MCM and ABS, contributes to the overall wire length reduction.

Fleet-Wide Architecture Comparison — Legacy vs Zonal

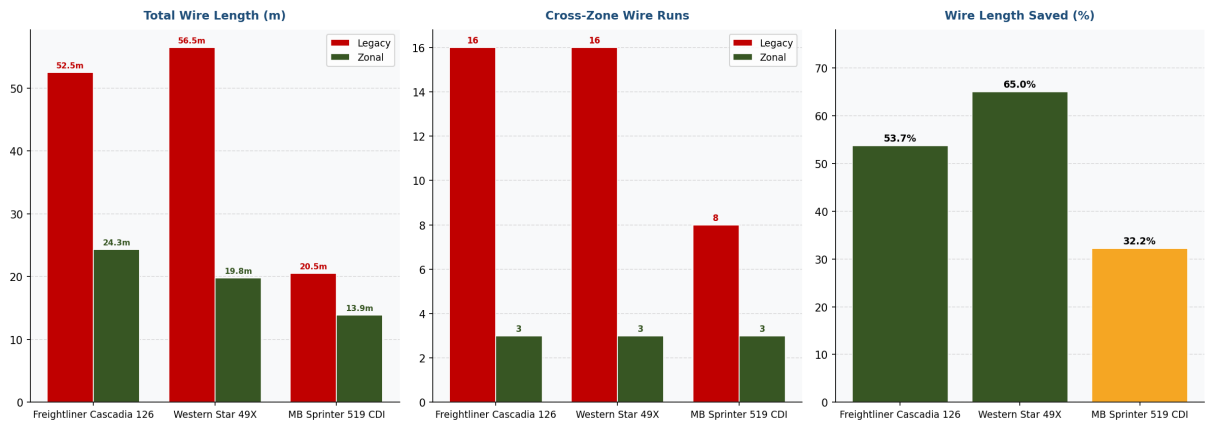


Figure A: Fleet-wide comparison. Wire length (left), cross-zone runs (centre), percentage savings (right).

7. Fleet-Wide Comparative Analysis

Analysis across three vehicles reveals how zonal architecture benefit scales with vehicle size and complexity.

Vehicle	ECUs	Connections	Legacy Wire (m)	Zonal Wire (m)	Wire Saved	Weight Saved	Zone Efficiency
Freightliner Cascadia 126	22	25	52.5m	24.3m	53.7%	3.4kg	7.0%
Western Star 49X	17	22	56.5m	19.8m	65.0%	4.4kg	-4.9%
Mercedes-Benz Sprinter 519 CDI	12	13	20.5m	13.9m	32.2%	0.8kg	31.2%

Table 10: Fleet-wide comparison.

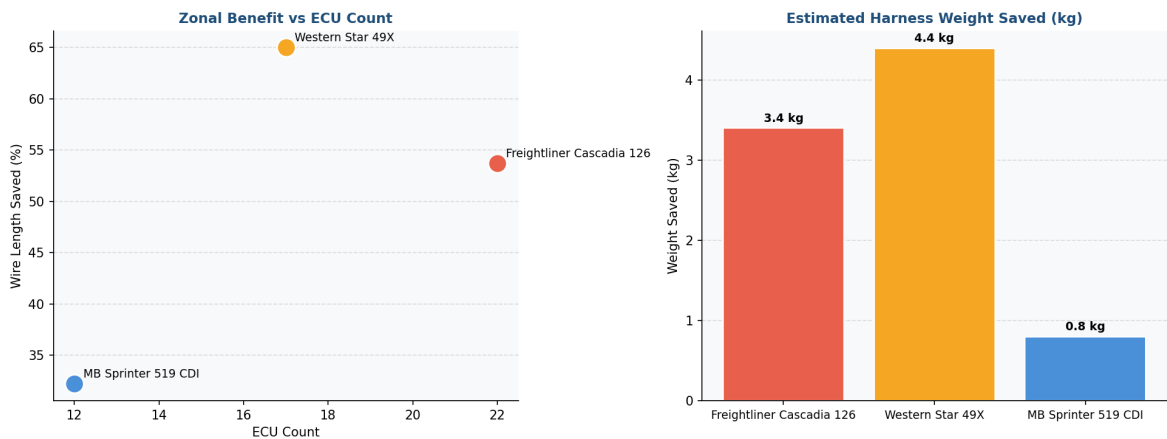


Figure C: Wire saved vs ECU count (left). Harness weight saved (right).

The Western Star 49X benefits most from zonal architecture, with a 65.0% wire reduction, despite having fewer ECUs (17) than the Freightliner Cascadia 126 (22). The ECU count threshold observation suggests that vehicles with fewer ECUs may benefit more from zonal architecture. Comparing zone efficiency scores, the Mercedes-Benz Sprinter 519 CDI has a significantly higher score (31.2%) than the Freightliner Cascadia 126 (7.0%) and the Western Star 49X (-4.9%), indicating varying degrees of optimization in zone boundary definition.

7.1 Key Cross-Fleet Findings

- Finding 1:** Zonal wire savings do not scale linearly with ECU count. Physical spread of ECUs matters more than total count.
- Finding 2:** Cross-zone wire runs always collapse to exactly 3 backbone segments in a 4-zone design — a structural property.
- Finding 3:** Zone efficiency scores are consistently low across all vehicles — the physical-vs-communication tradeoff is fundamental to automotive E/E design.
- Finding 4:** The Sprinter van (12 ECUs) saves 32.2% but the zone controller overhead is proportionally higher — suggesting a practical ROI threshold of ~15+ ECUs.

8. AI-Assisted Engineering Review

Generated by Groq Llama 3.3 70B from computed data. AI was instructed to act as a senior automotive E/E architect.

8.1 Design Recommendations

1. The MCM should be moved to the Chassis Zone (CHZ) to reduce its connections to the Powertrain Zone (PTZ) and improve zone efficiency.
2. The zone controller for the Cab Zone (CBZ) should be placed within 1.5 meters of the ICU to minimize connection lengths and reduce the load on the ICU.
3. A redundant communication path should be implemented for the ICU to ensure continued communication in the event of an ICU failure, such as duplicating critical connections through the CPC.

8.2 Analysis Limitations

- The analysis assumes a static zone layout and does not account for potential future changes in ECU connections or zone boundaries.
- The model does not consider the impact of zonal architecture on other factors, such as system latency, data throughput, or fault tolerance, which may affect the overall system performance and reliability.

9. Methodology and Data Sources

9.1 How the Tool Works

Step 1 — Input (YAML config): ECUs, zone assignments, connections, and zone distances described in a YAML text file.

Step 2 — Graph building (graph.py): Two NetworkX graphs: legacy (ECU-to-ECU, weighted by wire length) and zonal (ECU-to-ZC stubs + backbone).

Step 3 — Metrics (metrics.py): Wire length, cross-zone run count, harness weight computed by summing edge weights.

Step 4 — Zone optimality (optimizer.py): Greedy Modularity Community Detection finds communication-optimal zone groupings.

Step 5 — Three-way optimizer (constrained_optimizer.py): Greedy iterative reassignment at three modes: physical, combined (alpha=0.5), communication. Pareto curve computed across modes.

Step 6 — Fleet comparison (fleet_compare.py): Steps 1-4 run for all three vehicle configs.

Step 7 — AI review (reviewer.py): Computed data passed to Groq Llama 3.3 70B with structured automotive engineering prompt.

Step 8 — PDF generation (build_pdf_report.py): Matplotlib generates all diagrams. ReportLab assembles 18+ page PDF.

9.2 Data Sources

1. DTNA SS-1033423 — J-1939 Fault Code Source Address Descriptions. Daimler Trucks North America. NHTSA public database.
2. DTNA STI-503 — NGC sSAM, VPDM & BCA Wall Chart. Rev. Q, March 2020.
3. Western Star Bodybuilder Manual Rev 3.1.
4. Mercedes-Benz XENTRY public diagnostic documentation.
5. Park C, Cui C, Park S. Sensors. 2024;24(10):3248. DOI: 10.3390/s24103248.
6. SAE J1939 — Serial Control and Communications Heavy Duty Vehicle Network.

9.3 Key Assumptions

- Wire lengths estimated from published chassis dimensions: Cascadia 126 ~8.5m, Western Star 49X ~7.5m, Sprinter ~5.5m
- Harness weight uses 120 g/m bundled average for signal-wire harnesses (conservative)
- Analysis covers signal and data wires only — power distribution excluded
- Static ECU configuration — optional equipment not modelled
- Three-way optimizer wire threshold: 3.0m (adjacent-zone range for combined mode)
- All ECU data from public DTNA and Mercedes-Benz documents — no proprietary OEM data

Zonal E/E Architecture Analyzer | Open-source | <https://github.com/Rish-736/zonal-ee-analyzer>

Built alongside DTICI internship research — Rishit, ECE '28, VIT Vellore — June 2026